

Qualitative Field Tests

When is Jar, Stick, and Gloved Hand Test Performed?

Jar, stick, and gloved hand tests are an inexpensive way to distinguish between a potentially toxic freshwater harmful algal bloom (FHAB) and a non-toxic algal bloom. These tests are often performed after visual monitoring has identified a potential bloom. The simple tests can help differentiate between green algae and a FHAB and determine if sampling for cyanotoxins is advisable.

Who can use these field tests?

Jar and stick tests are simple experiments that most people can conduct. The supplies needed can be found in the field (a stick) or at home (a mason jar) and results are easy to interpret.

Steps to Jar, Stick, and Gloved Hand Test

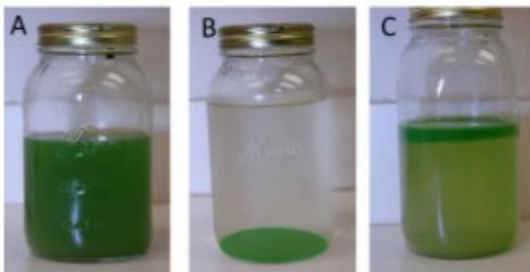
See the short, linked document and video below to learn how to conduct a jar, stick, and gloved hand test.



Simple, no-cost tests for blue-green algae
Use a jar or stick to check for blue-green algae in your lake or pond.

Adapted with permission from the Kansas Department of Health and Environment
A couple simple and free tests can help to determine if a lake has a blue-green algae community. The Minnesota Pollution Control

Note: No test for blue-green algae is perfect, including these. The jar test relies on the buoyancy of most free-floating blue-green algae. In Minnesota, most lake water conditions are the result of the seasonal



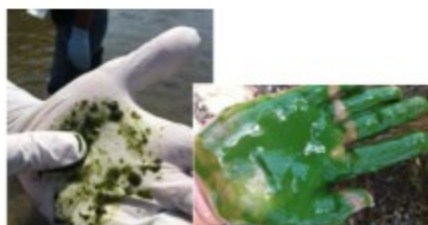
Jar Test

The jar test takes advantage of cyanobacteria's ability to regulate buoyancy and helps distinguish these organisms from common aquatic biota. Water containing the suspected FHAB is captured in a glass jar or other clear container and allowed to rest for 15 to 30 minutes. Cyanobacteria typically float to the surface or stay suspended in the jar, while other algae and sediment sink to the bottom.



Stick Test

The stick test can be used to separate planktonic cyanobacteria from common filamentous green algae. Planktonic FHABs often have an appearance like spilled paint on the water and will coat a stick or other item when it is dipped into the bloom. Filamentous green algae, which is nonharmful, can form stringy tangles just under the water surface or floating accumulations. They rarely coat the stick and will easily lift out of the water in long, hair-like strands. Benthic cyanobacteria may also lift easily out of the water but typically do not look like hair.



Gloved Hand Test

The gloved hand test can also be used to separate planktonic cyanobacteria from common filamentous green algae. Planktonic FHABs often do not have a lot of structure to them, so when you rub them together in the palm of your hand, they will break apart or smear. Filamentous green algae will have a structure containing long filaments and will often roll up in your hand.